real experiments, the “roulette balls” are photons, traveling at the speed of light, and the “roulette wheels” are polarizers that can be very far apart—in some experiments, as far as hundreds of kilometers. No light-speed signal sent after one photon arrives at a polarizer could possibly reach the other one before it too reaches a polarizer and makes its decision about what to do. In short, the results of real experiments with entangled photons mean that something, some influence, is going faster than light. Entanglement is not just an artifact of the mathematics of quantum physics: it’s a real phenomenon, an actual instantaneous connection between far-distant objects.

This is an astonishing result. How can it be right? What story of the world could account for this? The most obvious answer is: a nonlocal one. Bohm’s pilot-wave interpretation of quantum physics has no trouble at all with Bell’s theorem, because Bohm’s theory is explicitly nonlocal. This turns one of the apparent weaknesses of pilot-wave theory—its instantaneous connections between particles separated by huge distances—into a strength. Bell’s theorem strongly suggests that quantum physics must be nonlocal; pilot-wave theory merely makes this strange quantum behavior so obvious that we can’t ignore it.

But the cost of nonlocality is high. Relativity is one of the best-tested and most solid foundations of modern physics; nonlocality would put it at risk. Is there any other way out of Bell’s theorem? Is locality really the only assumption? Well, Gillian and Fatima did assume that their notes on what happened at the casino were a complete record of everything that happened there. Specifically, they assumed that there was only one outcome for each spin of a roulette wheel, the outcome they wrote in their notes. If there is somehow more than one outcome for each spin of a roulette wheel, each time a photon hits a polarizer, Bell’s proof falls apart. And this is exactly what happens in Everett’s many-worlds interpretation. According to Everett, each spin of a roulette wheel leads to both possible outcomes, red and black, branching in the many worlds of the universal wave function. So Bell’s theorem does suggest that the strangest piece of Everett’s scheme could be a necessary feature of the world if we don’t want to abandon locality.

Now we have two assumptions: locality and living in a single universe. One of them must be wrong, since Bell’s inequality is violated in real